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From Kratom to Semi-Synthetic Opioids: The Rise and Risks of MGM-15

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ABSTRACT

Kratom (*Mitragyna speciosa*), a plant native to Southeast Asia, has long been used for its stimulant and analgesic properties. 7-Hydroxymitragynine (7-HMG) is a potent and selective opioid agonist in vitro and demonstrates a potent opioid effect in living subjects, reversible by naloxone. It has been semi-synthesized into products that are readily available in retail and virtual shops. It is known that 7-HMG has earned the nickname "legal morphine," and has gained popularity among users seeking pain relief and/or a "high" comparable with prescription opioids. Medicinal chemistry efforts have led to synthetic 7-HMG derivatives such as MGM-15, where stereospecific saturation of the imine N(1)–C(2) double bond increases opioid receptor affinity and activity. Despite its higher in vitro opioid potency, MGM-15 is currently sold in the US for human consumption as a "research chemical" in tablet form, even though there is an absence of this being studied in humans and obviously no FDA approval. In this study, we analyzed commercially available MGM-labeled tablets using UPLC–MS/MS and subsequently evaluated the binding affinities of purified MGM-15 across multiple opioid receptors. Tablets contained an average of 10.9 ± 0.2 mg (10.7 to 11.2 mg) of MGM-15, with no naturally occurring kratom alkaloids or illicit substances detected. MGM-15 shows greater hMOR and hDOR binding affinities than 7-HMG, indicating the potential for higher opioid effects and risks, emphasizing the urgent need for more research to advise regulation and hopefully prevent misuse and harm.

1 | Introduction

The market for kratom (*Mitragyna speciosa* (Korth.) Havil.) is growing at a tremendous rate [1]. Freshly picked leaves have been traditionally used for their ethnopharmacological properties, including analgesic, antipyretic, antidiabetic, antidiarrheal, antitussive, antidepressant, and anxiolytic effects [2]. Kratom displays a polypharmacological profile, acting as a stimulant at low doses (2–4 g) and producing opioid-like effects at higher doses (5–8 g). This paradoxical dual activity makes kratom a unique and complex natural product of interest for pain management as well as for mitigating opioid addiction and withdrawal symptoms [3, 4]. Mitragynine (MTG) (Figure 1), the most abundant alkaloid found in native kratom plants, has long been believed to be responsible for many of the beneficial effects of kratom [5]. *M. speciosa* is widely used as a self-treatment for opioid use disorders, as a low-cost substitution/replacement for opioids, opioid withdrawal, and use disorders to other substances [4, 6]. As kratom products have gained popularity in the Western world, there has been

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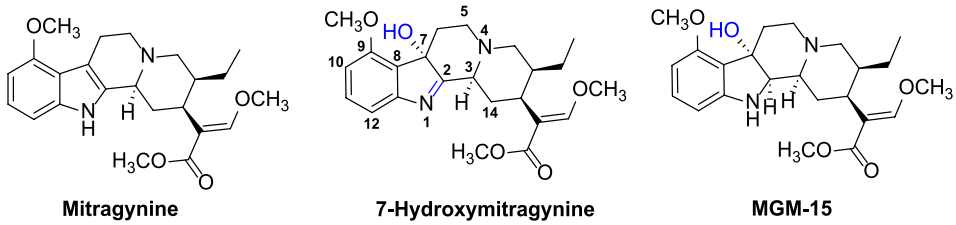


FIGURE 1 | Chemical structure of mitragynine, 7-hydroxymitragynine, and MGM-15.

a growing effort to explore the pharmacology of the alkaloidal constituents more thoroughly, including in vivo metabolites and semi-synthetic derivatives of the naturally occurring alkaloids.

7-Hydroxymitragynine (7-HMG) is an active metabolite of MTG [7]. 7-HMG is a potent opioid that exhibits significantly higher affinity for human μ-opioid receptors (hMOR) than MTG. In rat hot plate assays, 7-HMG demonstrated stronger analgesic effects than morphine, with an ED_{50} of 1.9 mg/kg versus 5.1 mg/kg for morphine, indicating its full agonist activity in living subjects. Unlike MTG, which reportedly acts as a biased partial agonist at hMOR, 7-HMG induces classical opioid side effects such as respiratory depression, tolerance, and cross-tolerance to morphine [8, 9]. 7-HMG is readily self-administered in rats and has demonstrated high abuse potential, with reinforcing effects mediated by hMOR. It has been shown that prior exposure to 7-HMG increases subsequent morphine use, suggesting it may enhance vulnerability to other opioids or act as a gateway drug. In contrast, MTG is not self-administered and reduces morphine intake, suggesting low abuse potential and possible therapeutic use [8]. Several 7-HMG-containing products are commercially available in the US market, but these potentially harmful products should not be classified as kratom. They are semi-synthetic derivatives produced through chemical oxidation of MTG-rich kratom extracts and have not undergone safety or efficacy evaluation in animals or humans [10]. In 2014, continuing the work of others who had previously modified and optimized 7-HMG derivatives, a Japanese research team reported the discovery and characterization of two novel 7-HMG analogs: MGM-15 and MGM-16. MGM-15 (Figure 1) was synthesized by reducing 7-HMG with $NaBH_4$ while MGM-16 was prepared similarly from 10-fluoro-7-HMG [11].

MGM-15 exhibited antinociceptive potency approximately 15- and 50-fold greater than morphine following subcutaneous and oral administration in mice, respectively. Similarly, MGM-16 demonstrated even higher potency, being approximately 71- and 240-fold more effective than morphine following subcutaneous and oral administration, respectively. Receptor binding studies revealed that MGM-16 has a high affinity for MOR and DOR, with receptor affinity (K_i) values of 2.1 and 7.0 nM, respectively, while MGM-15 showed K_i values of 6.4 nM (MOR) and 16 nM (DOR), indicating strong interaction with both receptor subtypes. These K_i values were derived from assays using membrane proteins isolated from rodent and pig brain tissue. MGM-16, a potent μ/δ-opioid dual agonist, showed superior efficacy to MGM-15 in vitro and in vivo, producing dose-dependent antiallodynic effects in sciatic nerve–ligated mice through both subcutaneous (0.1–0.4 mg/kg) and oral (0.5–2 mg/kg) administration, with peak effects at 15 to 30 min and full reversal of allodynia at 0.2 mg/kg (subcutaneous) and 1 mg/kg (oral), comparable with gabapentin (100 mg/kg). These derivatives were reported to be not only the most potent MOR agonists among all known MTG derivatives but also acted as full agonists at DOR [11]. This dual agonist activity highlighted the potential of these derivatives to treat acute, chronic, and neuropathic pain, as hDOR agonism has primarily been investigated for managing chronic pain rather than acute pain [12]. However, the targeting of DOR agonists has been abandoned by the pharmaceutical industry due to the high incidence of seizure occurrence [13].

Since these findings were published, no other primary research has been reported on these two novel derivatives. However, the synthesis of MGM-16 involves a complex, multistep process making large-scale production challenging compared to MGM-15 [14]. Conversely, MGM-15 is available for sale on multiple websites, typically marketed as "research purposes only" and accompanied by disclaimers stating the products are "not for human or animal consumption." Despite these warnings, many of these products are offered in a range of flavors and forms, including isolate powders, tablets, and liquid shots, offering customers multiple options to choose from. Currently, just a handful of vendors are selling MGM-15 products, most of which are labeled as "MGM." Increasing positive reviews on vendor websites and discussion about these products on online forums will inevitably lead to an increased demand for these products, similar to how 7-HMG products gained popularity. The aim of this study is to raise awareness about the availability of MGM products on the market and to conduct both qualitative and quantitative analyses to confirm the presence of MGM-15 and identify any other kratom alkaloids or potential adulterants followed by evaluation of their binding affinity at various receptors.

2 | Material and Methods

2.1 | Chemicals and Reagents

Tablets were purchased online from Ritually Pure (Niagara Falls, NY, USA). 7-HMG was synthesized from MTG as reported earlier [15]. Kratom alkaloids (MTG, mitraphylline, corynantheidine, paynantheine, corynoxine, corynoxine B, mitraciliatine, speciociliatine, and speciogynine) were isolated and purified in-house from a kratom alkaloid-rich extract as reported earlier [16]. Phenacetin was purchased from Sigma-Aldrich (St. Louis, MO, USA). LC–MS-grade acetonitrile, formic acid, acetic acid, ammonium acetate, and methanol were procured from Fisher Scientific (Fair Lawn, NJ, USA).

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2.2 | Experimental Procedure for the Synthesis of MGM-15

To a stirred solution of 7-HMG (150 mg, 0.36 mmol) in dry MeOH (6 mL), $NaBH_4$ (16.4 mg, 0.43 mmol) was added at 0°C under argon atmosphere. The reaction mixture stirred for 30 min (as monitored by TLC), H_2O was added to the reaction mixture followed by saturated aqueous $NaHCO_3$ solution (5 mL). The mixture was concentrated under reduced pressure; the crude residue was extracted with dichloromethane (3 × 30 mL). The combined extract was washed with brine, dried over $MgSO_4$, evaporated, and purified using flash column chromatography (gradient: 0%–40% EtOAc in *n*-hexanes), yielding 117 mg of MGM-15 (yield 78%) as a yellow solid (Figure 2). Previous literature report matches the NMR data (Figures S1 and S2) [11].

2.3 | Instrumentation and Chromatographic Conditions

Chromatographic analysis was carried out using a Waters Acquity I-Class Plus ultra-performance liquid chromatography (UPLC) system (Milford, MA, USA) coupled with a Xevo TQ-S Micro triple quadrupole mass spectrometer (MS/MS, Waters Corporation, Milford, MA, USA). Mass spectrometric detection utilized electrospray ionization (ESI) operated in positive ion mode. The target analytes were quantified using multiple reaction monitoring (MRM), ensuring high selectivity through the monitoring of specific precursor-to-product ion transitions. Instrument parameters, including capillary voltage, desolvation temperature, cone gas flow, and desolvation gas flow, were optimized to improve signal strength and detection sensitivity. The MRM transitions for MGM-15 and kratom alkaloids are presented in Table S1. Parent to product ion mass spectra of MGM-15 are presented in Figure S3.

Chromatographic separation of MGM-15 and kratom alkaloids was achieved using a Waters Acquity BEH C18 column (1.7 µm, 2.1 × 100 mm) with gradient elution. The mobile phases consisted of aqueous 2.5 mM ammonium acetate buffer (pH 3.5) as solvent A and acetonitrile as solvent B. The optimized gradient elution program was as follows: 0–3 min at 30% B; 3–3.9 min, gradient from 30% to 40% B; 3.9–4.0 min at 40% to 70% B; 4.0–4.5 min at 70%; 4.5–4.6 min at 70% to 30% B; and 4.6–5.5 min at 30% B. The flow rate was maintained at 0.35 mL/min, the column temperature was set to 55°C, and the autosampler was kept at 10°C. The retention times of MGM-15 and the internal standard (IS, phenacetin) were 1.14 and 1.66 min, respectively. The injection volume used was 2 µL. The strong needle wash solution comprised water, acetonitrile, methanol, and isopropanol (1:1:1:1, v/v/v/v) with 0.1% formic acid. The weak needle wash solution consisted of acetonitrile, methanol, and water (1:1:2, v/v/v) with 0.1% formic acid [17].

2.4 | Preparation of Standard and Quality Control Samples

Stock solutions (1 mg/mL) of MGM-15 and kratom alkaloids were prepared in methanol, followed by the preparation of secondary stocks (50 µg/mL) and spiking stocks, which were further used to prepare calibration standards (CS) in the concentration range of 1 to 500 ng/mL for each analyte. Quality control (QC) samples were prepared similarly at four concentration levels: 1.0 ng/mL [lower limit of quantification (LLOQ)], 3.0 ng/mL [low quality control (LQC)], 150 ng/mL [medium quality control (MQC)], and 450 ng/mL [high quality control (HQC)]. Fresh CS and QC samples were prepared by adding 2.5 µL of spiking mix stock solution to 22.5 µL of blank methanol, and then vortex-mixed for 5 min using a BenchMixer (Benchmark, San Francisco, CA, USA). Methanol:water mixture (80:20, 100 µL) containing 5 ng/mL of IS was added to 25 µL of sample and vortexed for 5 min. These were filtered using a Multiscreen Solvinert 96-well 0.45-µm filter plate (Millipore, St. Louis, MO, USA) followed by centrifugation at 841 *g* for 5 min at 4°C and injected onto the LC–MS/MS for analysis. Data acquisition and processing were performed using MassLynx version 4.2.

2.5 | Method Validation for Quantitation of MGM-15

A UPLC-MS/MS method was partially validated for the quantification of MGM-15 in tablet formulations, following the guidelines outlined in the USFDA's "Analytical Procedures and Methods Validation for Drugs and Biologics" [17, 18]. Key validation parameters assessed included specificity, limit of detection (LOD), LLOQ, linearity, accuracy, and precision. Specificity was evaluated to ensure that MGM-15 could be accurately identified and quantified in the presence of other expected components or potential interferences. The LOD and LLOQ were established based on the signal-to-noise ratio and the consistency of repeated injections. To assess linearity, a calibration curve was plotted across the experimental range, and linear regression analysis was performed. Accuracy and precision were tested at four concentration levels: LLOQ, LQC, MQC, and HQC. Intra-day accuracy and precision were determined using six replicates per level ($n = 6$) within a single day, while inter-day performance

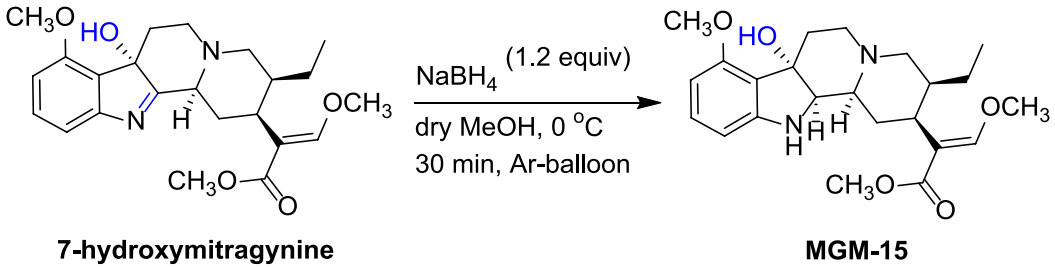


FIGURE 2 | Synthesis scheme of MGM-15 from 7-hydroxtmitragynine.

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was assessed over three consecutive days using the same number of replicates. Precision was calculated as the percent relative standard deviation (%RSD).

2.6 | Quantitative Analysis of Commercially Available MGM Tablet

A total of three tablets were used for analysis with an average weight of 456.8 ± 3.6 mg. These were finely ground using a mortar and pestle. The resulting powder was transferred to a 100-mL volumetric flask and dissolved in methanol. It was further sonicated for 60 min to ensure complete extraction, followed by filtration. The filtrate was diluted 100,000-fold with methanol to fall within the linearity range, processed as described in Section 2.4, and analyzed using the UPLC-MS/MS system. These tablets, along with a sample of kratom leaf aqueous extract and a commercially available kratom product (Optimized Plant Mediated Solutions Gold capsules, Choice Organics, Los Angeles, CA, USA) were also analyzed for further confirmation if any phytoconstituent interfered with the analysis of MGM-15 [17, 19, 20].

These commercially available MGM-15 tablets were screened for illicit substances, prescription drugs, and cannabinoids using a gradient UPLC-MS/MS method following previously reported methods [19, 20]. A few of the target compounds included tianeptine, hydromorphone, morphine, midazolam, oxycodone, fentanyl, tramadol, cannabidiol, Δ9-tetrahydrocannabinol, buprenorphine, and a complete list of analytes is available in Table S2. This method was not validated for quantitative analysis but was used for targeted screening to identify compounds with mass transitions matching those of known illicit drugs, prescription medications, and/or cannabinoids. If any target analyte had been detected in an MGM tablet, a separate validated quantitative analysis would have been performed.

2.7 | In Vitro Receptor Binding Assays

The binding assay was performed using monoclonal opioid receptors expressed in Chinese Hamster Ovary (CHO) cell lines. MGM-15, 7-HMG, and MTG were screened for in vitro binding affinity at hMOR, hDOR, and human κ-opioid receptors (hKOR), as previously reported with some modifications [21]. [^{3}H] DAMGO, [^{3}H] U69593, and [^{3}H] DADLE radioligands were employed to label hMOR-CHO, hKOR-CHO, and hDOR-CHO, respectively. Saturation binding assays were conducted to determine the K_d and B_{max} of each radioligand at its corresponding receptor. Prior to these studies, membrane evaluations were conducted to identify the optimal protein concentration for each receptor. Briefly, 20 μg of protein was incubated with a radioligand in the presence or absence of compounds in an assay buffer (50-mM Tris, 3-mM $MgCl_2$, and 0.2-mM EGTA, pH 7.4). Specific (i.e., opioid receptor-related) binding at these opioid receptors was determined as the difference between total and nonspecific binding (NSB). NSB was determined using 10 μM of appropriate receptor-selective competitors: naltrexone for hMOR, U69593 for hKOR, and SNC80 for hDOR. Following incubation for 90 min at 30°C, the bound and unbound radioligands were rapidly separated by filtration under vacuum through glass fiber filters pre-soaked with 0.05% polyethyleneimine (PEI) and rinsed several times with an ice-cold washing buffer (50-mM Tris) using a 96-well plate cell harvester. The filters were dried for 20 min at 50°C and counted for radioactivity using a Microbeta2 PerkinElmer microplate reader (Ramsey, MN, USA) with a scintillation cocktail. The receptor binding assays were conducted in triplicate (two plates) and repeated at least twice on different days. Displacement binding curves were analyzed using nonlinear least-squares regression (GraphPad Prism, GraphPad Software, San Diego, CA, USA) to determine IC_{50} (concentration causing a half-maximal inhibition of control-specific binding) values, which were subsequently converted to K_i values using the Cheng–Prusoff equation.

3 | Results and Discussion

Compound parameters for MGM-15 were added to a previously reported UPLC–MS/MS-based analytical method available for the simultaneous quantification of kratom alkaloids and partially validated for MGM-15 [17]. No interference of kratom alkaloids was observed at the retention time of MGM-15 and vice versa. LOD was defined as the lowest concentration of MGM-15 spiked in a methanol–water mixture that produced a signal-to-noise (S/N) ratio of ≥3:1. LLOQ was established through a series of preliminary experiments, confirming the method's high sensitivity. At an LLOQ of 1 ng/mL, the signal-to-noise ratio exceeded the recommended threshold of 10:1, and both accuracy and precision met the acceptance criteria—within 20% of the nominal concentration and % relative standard deviation (RSD), respectively. The calibration curve was linear over the concentration range of 1 to 500 ng/mL, with correlation coefficients consistently greater than 0.99 in all analytical runs. The method's reliability and reproducibility were further validated through intra-day and inter-day assessments of accuracy and precision. For the LQC, MQC, and HQC, accuracy and precision (%RSD) remained within 15% of the nominal values. For the LLOQ, which allows slightly broader limits due to its low concentration, both parameters were within 20% (Table S3). These findings confirm that the method provides consistent, accurate, and precise quantification of MGM-15 across a wide range of concentrations and over multiple days. The analysis of commercially available MGM tablets confirmed the presence of MGM-15 and the absence of other kratom alkaloids or known adulterants, including 7-HMG. Each tablet contained 10.9 ± 0.2 mg of MGM-15. The representative chromatograms of MGM-15 for both reference standard and tablet product are presented in Figure S4. Screening for illicit substances, prescription drugs, cannabinoids, and kratom alkaloids revealed no contamination or substitution, suggesting that these tablets are chemically consistent with their labels.

A receptor binding assay was further employed to evaluate the affinity of MGM-15 at opioid receptors utilizing CHO recombinant cell lines expressing hMOR, hDOR, and hKOR and compared with MTG and 7-HMG. In contrast, Matsumoto et al. conducted their binding studies using guinea pig tissue homogenate membranes for MOR and DOR, and male Sprague–Dawley rat brain homogenate membranes for KOR [11]. These membranes express all receptors with multiple binding sites and enzymes that could interact with the radioligand or affect the

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TABLE 1 | Affinities ($K_i \pm$ SEM) of mitragynine, 7-hydroxymitragynine, and MGM-15 at opioid receptor subtypes.

Compound	hMOR (nM)	hKOR (nM)	hDOR (nM)
Mitragynine	299 ± 15	248 ± 45	2009 ± 281
7-Hydroxymitragynine	72 ± 11	136 ± 21	202 ± 51
MGM-15	28 ± 7	340 ± 79	59 ± 4

Abbreviations: hDOR, human δ-opioid receptor; hKOR, human κ-opioid receptor; hMOR, human μ-opioid receptor.

assay. Because the objective of this study was to determine if MGM-15 poses a potential risk to human health, binding affinities at opioid receptors were determined in human-based transfected cells expressing individual receptor subtypes.

7-HMG is itself a concern to public health due to its high potency at the hMOR, contributing to opioid-associated risks such as sedation, respiratory depression, and potential for abuse and dependence. However, MGM-15 could represent an even greater threat. MGM-15 binds to hMOR with a K_i of 28 ± 7 nM, suggesting it to be approximately 2.6-fold more potent than 7-HMG (Table 1). At hDOR, MGM-15 displays a K_i of 59 ± 4 nM, 3.4-fold higher affinity than 7-HMG. Although MGM-15 is slightly weaker at the hKOR with a K_i of 340 ± 79 nM compared with 7-HMG (136 ± 21 nM), its overall binding profile across multiple opioid receptors suggests a strong opioid-like effect. Although MTG, the main alkaloid in traditional kratom products, is a weak agonist at opioid receptors, approximately 4.2- and 10.7-fold weaker at hMOR, 10- and 34-fold weaker at hDOR, and 1.8-fold weaker than 7-HMG but 1.4-fold stronger than MGM-15 at hKOR, respectively. This indicates that MTG is a relatively weak partial agonist in vitro at these receptors and contributes to a significantly milder opioid effect profile compared to either 7-HMG or MGM-15.

Despite these pharmacodynamic properties, MGM-15 has not undergone FDA review for safety, efficacy, or toxicity in humans. The presence of MGM-15 in flavored, consumer-ready formats despite being labeled as "not for human consumption" suggests a loophole in current regulatory frameworks. With its high affinity for both hMOR and hDOR and increasing availability, MGM-15 poses a potential for misuse and public health risk comparable to, or even greater than, 7-HMG. FDA has already issued warning letters for products containing 7-HMG due to its pharmacological profile and potential for harm, similar regulatory scrutiny is warranted for MGM-15.

4 | Conclusions

In recent years, legislation and regulations on products containing kratom and their constituents have become increasingly more stringent. Kratom products, including 7-HMG, are completely banned in six US states and regulated in several others. Moreover, on July 15, 2025, the US Food and Drug Administration (FDA) issued warning letters to seven companies for the unlawful marketing of products containing 7-HMG. Recently, the FDA has initiated a significant move to safeguard public health by recommending the scheduling of 7-HMG products under the Controlled Substances Act, aiming to curb the risks posed by dangerous, illegal opioids [22]. This action highlights the FDA's increasing concern over the sale of powerful, novel opioid products being marketed to US consumers, both online and through retail outlets, such as smoke shops, gas stations, and convenience stores. Due to the decreasing accessibility of kratom and 7-HMG, vendors have found that newer derivatives may be exempt from these laws and regulations, allowing for the marketing and sale of products such as MGM-15.

MGM-15 is a potent, unregulated opioid compound now found in consumer products despite lacking FDA approval or safety data in humans, and this is the first time analysis of a consumer product containing this compound. Its increasing availability highlights the urgent need for regulatory oversight and market removal to protect public health. The potential Schedule 1 classification of 7-HMG by the FDA leaves an open door for MGM products to fill the void. Food products are required to have premarket approval by the FDA to ensure their safety, and MGM-15, along with other synthetic derivatives of kratom alkaloids (i.e., 7-HMG and mitragynine pseudoindoxyl), represents potentially highly dangerous products that would be recognized as unsafe under section 409(a) of the FD&C Act.

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Conflicts of Interest

The authors declare no conflicts of interest. C.R.M. is an expert witness as a plaintiff and defendant in wrongful death kratom civil lawsuits.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** dta3952-sup-0001-Supplementary_Material.docx. ^{1}H-NMR spectrum of methyl (*E*)-2-((2*S*,3*S*,7a*S*,12b*S*)-3-ethyl-7a-hydroxy-8-methoxy-1,2,3,4,6,7,7*a*,12,12*a*,12*b*-decahydroindolo[2,3-*a*]quinolizin-2-yl)-3-methoxyacrylate (MGM-15). **Figure S2:** dta3952-sup-0001-Supplementary_Material.docx. ^{13}C-NMR spectrum of (*E*)-2-((2*S*,3*S*,7a*S*,12b*S*)-3-ethyl-7a-hydroxy-8-methoxy-1,2,3,4,6,7,7*a*,12,12*a*,12*b*-decahydroindolo[2,3-*a*]quinolizin-2-yl)-3-methoxyacrylate (MGM-15). **Figure S3:** Product-ion spectra of MGM-15. **Figure S4:** Representative MRM chromatograms of blank (A), at LLOQ level (1 ng/mL, B), commercially available tablet (C) and internal standard (D). **Table S1:** Mass spectrometer source parameters and retention times for MGM-15, kratom alkaloids and internal standard (IS). **Table S2:** Mass spectrometer source parameters for illicit compounds. **Table S3:** Intra-day and inter-day accuracy and precision data for the determination of MGM-15 using UPLC–MS/MS.

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